FLCR-10 - Temporal Windows And Hamiltonian Readouts

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper turns static carried centers into ordered window objects. A Hamiltonian readout is a finite sliding-window transform with source indices, source centers, forward receipts, backward receipts, and emitted composite centers. The paper closes finite window emergence and kappa-ledger ordering while keeping physical time and unbounded McKay/O2-prime exactness in later claim lanes.

Keywords

Hamiltonian window; sliding window; provenance; kappa ledger; temporal readout.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Temporal Windows And Hamiltonian Readouts**. The paper's immediate contribution is: Defines temporal windowing over transported centers.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Temporal Windows And Hamiltonian Readouts.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines Hamiltonian windows as provenance-preserving finite sliding-window transforms.
- Proves the count n - w + 1 for finite ordered center sequences.
- Separates receipt-level backward reconstruction from physical time reversal.
- Routes McKay threshold bands and unbounded parity to later bounded-to-unbounded evidence.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- `09_Hamiltonian_Window_Emergence.md`
- `virtuous/09_Hamiltonian_Window_Emergence_VIRTUOUS.md`
- `supplements/09_INTERNAL_REASONING_SUPPLEMENT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Hamiltonian window.** A contiguous width-w slice over an ordered sequence of center states.
- **Forward receipt.** The source centers recorded in source order.
- **Backward receipt.** The same source centers recorded in reverse order as an audit inverse.
- **Kappa ledger.** A monotone event scalar record using kappa = ln(phi)/16 as an accounting unit.

4. Formal Claims

Claim	Lane	Statement
Theorem 10.1	`standard_theorem_citation_bound_result`	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n.
Theorem 10.2	`receipt_bound_internal_result`	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center.
Protocol 10.3	`falsifier_or_open_obligation`	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 10.1	formal construction	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Theorem 10.2	finite/executable result	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 10.3	obligation/falsifier	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	`FLCR-10` defines or uses **Temporal Windows And Hamiltonian Readouts** as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Take an ordered finite list of carried centers.

2. Choose a width w and enumerate start indices 0 through n-w.
3. For each start, emit the contiguous source window and record forward and backward receipts.
4. Treat reversal of the backward receipt as audit reconstruction, not physical time reversal.

Proof Sketch

There are n-w+1 valid start positions for a width-w contiguous window inside a length-n sequence. Each emitted object is defined from its source indices, so provenance is preserved by construction. Recording the reverse order gives an audit inverse because reversing a finite list twice returns the original order.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is `Temporal Windows And Hamiltonian Readouts`. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

Recent External Quantum Evidence Intake

These rows add newly routed external physics papers as citation-bound or calibration-bound evidence. They are not CAM receipts unless a later tool run reconstructs their signatures.

Source	Lane	How It Helps This Paper	Boundary
EXT-FFS-REALIZATION-2026: Realization of fractional Fermi seas; DOI `10.48550/arXiv.2602.17657`; arXiv `2602.17657`	`external_calibration_result`	Provides an external example of a time-windowed preparation protocol where cyclic interaction ramps create a stable excited state with readable downstream signatures.	The paper may use this as a comparator for temporal-window and prestate/poststate reasoning, not as proof that LCR generated the state.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 09 legacy body	Hamiltonian windows, scalar sweep, kappa timeline, McKay threshold stack.
Virtuous Paper 09	Rich proof body, production counts, threshold bands, CAM/GLM crystal import.
Internal supplement 09	Sliding-window, provenance-transform, Lyapunov-style scalar, threshold-band registry.
CAM Claim 100 Score Audit	Paper 09 scored 84, bounded/system-closeable; unbounded McKay/O2-prime remains later-stage.

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GLM-OBLIGATION-CLOSURE	The GLM closure matrix independently compares verifier findings to named obligations and advances 11 obligations to partial closure while identifying the untouched set.	When fed back through the claim lanes, the matrix stops the suite from carrying stale open labels where bounded verifier evidence already exists, while preserving untouched obligations as active next channels.	Obligation status promotion becomes auditable instead of rhetorical.; Papers 09, 10, 13, 16, 18, and 33 gain bounded closure support from specific verifier findings.; The suite gains a clean split between partial closure, engineering scope, and untouched physical/calibration bridges.	Partial closure is not unconditional closure.; CKM, quark/color measurement, Higgs/QFT mass calibration, GR curvature bridges, and full Moonshine remain separate dependency lanes where the matrix says they remain open.
REAPPLY-RULE30-ADDRESS-HORIZON	The Rule-30 address horizon supplies a hashed one-megabit center-column reference, a named one-gigabit resource target, and a billion-bit address-space package.	When reapplied, the address horizon becomes a finite prediction and enumeration substrate: CA prediction, process hashing, paper-window addressing, and CAM projection can share a concrete bit-address carrier.	The CA prediction layer now has a finite, content-addressed reference rather than a purely conceptual Rule-30 mention.; Finite horizon claims can be stated strongly without pretending they prove infinite non-periodicity.; Rule-30 data can be used as an addressable carrier for recursive paper and process scans.	The finite reference does not prove infinite non-periodicity.; The one-gigabit Wolfram resource remains a named external acquisition/calibration lane if bit-for-bit reproduction is required.

Combined Recursive Effects

Combination	Result	Solves
COMBO-ADDRESSABLE-PUBLICATION-RUNNER	Rule-30 finite address data, CAM node routing, and paper-runner protocol combine into an addressable publication runner: paper windows and proof artifacts can be enumerated, hashed, routed, and replayed.	This closes the gap between a paper as text and a paper as a state-bound ribbon with executable receipts.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GLM-OBLIGATION-CLOSURE	`normal_form_result`	Adds a closure matrix that compares independent GLM verifier findings against open obligations and records bounded promotions instead of leaving the papers in their earlier unresolved posture. Evidence facts: 12 findings considered against 29 obligations; 11 obligations advanced to partial; 17 untouched; closed-bounded findings include GLM-F1-SM01, GLM-F2-SP002, GLM-F4-RULE30P3, GLM-F6-YM-EXT02, GLM-F7-SM04, GLM-F8-HODGE-EXT01.	Promotions are lane-bound and partial where the matrix says partial; external physical bridges remain separate calibration obligations.
CAP-RULE30-ADDRESS-HORIZON	`receipt_bound_internal_result`	Adds a concrete address-horizon layer for cellular-automaton evidence: a local one-megabit Rule-30 center-column reference, a named one-gigabit external resource, and a billion-bit address-space package. Evidence facts: Rule-30 local reference: 1000000 bits, 125000 bytes, sha256 4bf71a265b7a3b9847703f274c736c552825a859fd2251d8bfad75b1fe05b3e9; billion-bit address package: 1000000000 bits, 120 chunks, 339 process hashes.	Finite address evidence strengthens prediction and addressability claims; it does not by itself prove infinite non-periodicity or physical identity.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	`receipt_bound_internal_result`	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.510203.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
window/aggregate synthesis	`normal_form_result`	The window or aggregate action is closed as a source, receipt, and next-prestate assignment surface.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	`receipt_bound_internal_result`	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-10 cross-lift comparison	`cam_crystal_reapplication_result`	Family tower binds FLCR-10, FLCR-20, FLCR-30, FLCR-40; avg TarPit mass=0.510965; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
temporal window readout	`receipt_bound_internal_result`	Temporal windows are closed as replayable state readout surfaces.	External time observables require calibration.

Claim Posture After This Pass

`FLCR-10` should now be read as a resolved-state contribution for `Temporal Windows And Hamiltonian Readouts` at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- `FLCR-11` through `FLCR-18` for window, receipt, admission, CA, algebraic,

curvature, residue, and exceptional machinery.

- `FLCR-28` through `FLCR-30` for CAM/crystal, corpus ribbon, and universal

normal-form intake.

- `FLCR-31` through `FLCR-40` only after the LCR-native result has been cited

and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- Audit reversibility is not physical time reversal.
- Window exactness does not imply unbounded arithmetic extraction.
- Kappa ledger ordering is not physical energy calibration without later units and measurements.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): `09`/window_action.

Original slot	Family	Lift	Role	Proof form
`09`	window_action	0	order-1 slot-9: read the ribbon through windows, meta-readouts, or synthesis bands	window count, source preservation, and meta-ribbon proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 08 and reopened at original slot 09 (Hamiltonian Window Emergence).

It must produce: formal result of original slot 09 plus the same-family lift path toward slot 19.

Semantic transition: read the ribbon through temporal, observer, Hamiltonian, meta, or synthesis windows.

Accepted formalism targets to bind in the journal rewrite:

- windowed dynamical systems
- operator/readout notation
- Hamiltonian or energy-style accounting when explicitly bounded
- multi-scale dependency summaries

Slot	Family	Lift	Produced proof form
`09`	window_action	0	window count, source preservation, and meta-ribbon proof

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Past work reads across scales: local paper, same-family lift, 3/5/7/9 reasoning windows, and standards-window 2/4/8/16/32 windows.

Primary Evidence Inputs

Source	Placement reason
`09_Hamiltonian_Window_Emergence.md`	primary assigned source for the paper's formal spine
`supplements/09_INTERNAL_REASONING_SUPPLEMENT.md`	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations
`virtuous\09_Hamiltonian_Window_Emergence_VIRTUOUS.md`	prior enriched paper body; should be mined for mature claims and proof ordering

Secondary And Orbital Evidence Inputs

Source	Placement reason
`supplements/CRYSTAL_CAM_PROJECTOR_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/RECEIPT_VERIFIER_CATALOG.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_PYRAMID_INDEX.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md`	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
`CAM_CLAIM_100_SCORE_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_TOOL_CLOSURE_MATRIX.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`NSIT_REASONED_CLOSURE_PASS.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`ORBITAL_DATA_CROSS_POPULATION_AUDIT.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`AGENT_CRYSTAL_WORK_HARVEST.md`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
`PAPER_REWORKS_CRYSTAL_PROJECTION.json`	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

09	84	bounded/system-closeable	Hamiltonian window, Paper 05 accumulated carrier, Paper 10 readout, Paper 16 continuum links	unbounded McKay/O2-prime correction collapse

Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Moonshine/VOA Finite Sector Split	`receipt_bound_internal_result`	Promote finite VOA sector, centroid-chain, and windowed sector claims when the receipt is attached.	Attach centroid/VOA receipt, finite sector count, and theorem/citation route for any moonshine identification.	Full Moonshine identity, McKay parity, and unbounded identification remain theorem/data-bound.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Moonshine Voa Finite Sector	This paper may use moonshine/VOA language for finite sector or windowed representation surfaces only where locally supported.	Centroid/VOA receipt, finite sector chain, lattice/representation rows, and theorem/data route for stronger identities.	Promote finite VOA sector and centroid-chain claims when the finite receipt is attached.	Full Moonshine identity, McKay parity, and unbounded identification remain theorem/data-bound.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMPLX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-10.25: Paper 10.25 - Toolkit for the T10 Master Receipt	73	Paper 10.25 describes the tools for reviewing the T10 master receipt. These tools expose receipt binding, transport classification, local witness replay, and lookup cache materialization. They do not close the open li...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-11_UPGRADED: Paper 11 - Theory Admission Gate (UPGRADED: Affirmative Encoder-Bound Filter Physics Map)	69	Paper 11 **proves the theory admission gate** for the CQECMPLX suite. An external theory is not admitted because it sounds compatible with the framework, and it is not discarded because a first transport attempt fails...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-11: Paper 11 - Theory Admission Gate	69	Paper 11 proves the theory admission gate for the CQECMPLX suite. An external theory is not admitted because it sounds compatible with the framework, and it is not discarded because a first transport attempt fails. It...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-09.50: Paper 9.50 - Hamiltonian Window Claim Contract	69	Paper 9.50 lets later papers import Hamiltonian window emergence honestly. It preserves the finite temporal construction without letting physical time language outrun the receipt.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	14 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	`FLCR-10`
Legacy source slot(s)	`09`
Ribbon role	order-1 slot-9: read the ribbon through windows, meta-readouts, or synthesis bands
Proof form	window count, source preservation, and meta-ribbon proof
Received state	state emitted by prior slot 08 and reopened at original slot 09 (Hamiltonian Window Emergence)
Produced state	formal result of original slot 09 plus the same-family lift path toward slot 19

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 10.1	`standard_theorem_citation_bound_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n.
Theorem 10.2	`receipt_bound_internal_result`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center.
Protocol 10.3	`falsifier_or_open_obligation`	Receipt, source card, validator, citation, or CAM route named in the paper manifest	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone.

Figure Plan

Figure	Purpose	Caption
FLCR-10-F1	Windowed corpus readout	Schematic showing how `FLCR-10` turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-10-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-10-F3	Same-family lift context	Diagram placing this paper in the original `00-79` ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-10-F1: Windowed corpus readout

```
received state
  -> Windowed corpus readout
  -> formal claim lane
```

```
-> evidence object / receipt / citation
-> produced state
-> remaining residue
```

Table Plan

Table	Purpose
FLCR-10-T1	Window readout table
FLCR-10-T2	Claim-lane/evidence-boundary table
FLCR-10-T3	Archive evidence card and source-backed upgrade table
FLCR-10-T4	Workbook replay and falsifier table

Worked Example

Read the paper through local, same-family, 3/5/7/9, and 2/4/8/16/32 windows. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
multi-scale	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
readout	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
same-family lift	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.
window	Defined by this paper as part of the `window_action` proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-10-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?

Check	Reviewer question
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

`FLCR-10` (`Temporal Windows And Hamiltonian Readouts`) occupies the `window_action` role at lift depth `0`. The paper receives state emitted by prior slot 08 and reopened at original slot 09 (Hamiltonian Window Emergence). It produces formal result of original slot 09 plus the same-family lift path toward slot 19. The state transition is: read the ribbon through temporal, observer, Hamiltonian, meta, or synthesis windows.

The paper-local proof form is:

```
window count, source preservation, and meta-ribbon proof
```

The integration result is a window readout that preserves sources, closures, residues, and next-prestate assignments. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
windowed dynamical systems	Formal carrier for the paper-local result.
operator/readout notation	Formal carrier for the paper-local result.
Hamiltonian or energy-style accounting when explicitly bounded	Formal carrier for the paper-local result.
multi-scale dependency summaries	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 10.1	`standard_theorem_citation_bound_result`	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n.
Theorem 10.2	`receipt_bound_internal_result`	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center.
Protocol 10.3	`falsifier_or_open_obligation`	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	`09_Hamiltonian_Window_Emergence.md`, `virtuous/09_Hamiltonian_Window_Emergence_VIRTUOUS.md`, `supplements/09_INTERNAL_REASONING_SUPPLEMENT.md`	routed evidence
Evidence inputs	`CAM_CLAIM_100_SCORE_AUDIT.md`, `NSIT_TOOL_CLOSURE_MATRIX.md`, `NSIT_REASONED_CLOSURE_PASS.md`, `PAPER_REWORKS_CRYSTAL_PROJECTION.json`	routed evidence
CAM/crystal sources	`PAPER_REWORKS_CRYSTAL_PROJECTION.json`, `AGENT_CRYSTAL_WORK_HARVEST.json`, `runtime standards artifact`, `notebook-derived source bundle`	routed evidence
Standards-window inputs	`window_summary.json`, `node DBs`, `window DBs`	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting

the operation;

4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-10-F1: Windowed corpus readout	Visualizes the proof object, routing, or boundary.
FLCR-10-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-10-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-10-T1: Window readout table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-10-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-10-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-10-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.